

Wildfire Risk Prediction Using Vegetation Stress in Australia

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Abstract

Wildfires in Australia have intensified significantly in recent years, contributing to the loss of approximately 35% of the country’s forests between 2016 and 2021. Current wildfire warning systems primarily respond after fires ignite, limiting their ability to support early prevention and preparedness efforts. This research investigates whether pre-fire vegetation stress can be used to predict wildfire burn severity before ignition occurs. Using over two decades of satellite observations from NASA SMAP and ERA5-Land datasets, the study analyzes vegetation optical depth (VOD), soil moisture, and related environmental variables as indicators of vegetation drought and flammability. Historical wildfire records are combined with these satellite-derived features to train a machine learning model capable of classifying fires into low and high severity categories. Preliminary results show that the model achieves approximately 69% classification accuracy, 82% recall, and 75% F1 score for strong fires and 64% classification accuracy, 47% recall, and 54% F1 score for weak fires. The long-term goal of this work is to develop an interactive real-time mapping platform that visualizes vegetation stress and predicted wildfire risk zones to support earlier intervention and community preparedness.

1 Introduction

Australia is experiencing a wildfire crisis of escalating severity. Between 2016 and 2021, approximately 35% of the country’s forests burned, a scale of destruction that has led to more pressure on emergency management systems, displaced communities, and irreversible ecological damage [1]. Vegetation dryness is a partial cause of these large fires since as plant tissue loses moisture over weeks and months, fuels become more ignitable, fires spread faster, and containment becomes exponentially more difficult [2]. Therefore, the challenge is identifying where and when conditions may cause a catastrophic fire.

Current warning systems are not great for this task. Australia’s Fire Danger Rating System and analogous frameworks such as the U.S. Smokey Bear program operate at broad regional scales and get their risk estimates primarily from real time weather variables: temperature, wind speed, and relative humidity. While these systems provide value in communicating acute

danger, they are more reactive. They do not incorporate satellite-derived vegetation conditions, and they offer little information about the weeks-long drying process that determines how severely a fire will behave once it ignites. As a result, communities and fire agencies often receive inadequate warning before a low-danger day escalates into a catastrophic event [2].

Prior studies on fire risk modeling have relied predominantly on atmospheric and fuel-load variables observed at or near the time of the fire, rather than looking at trends over the prior weeks and months. Whether those longer duration data points can predict fire severity as well as current atmospheric data has not been completely tested.

This paper addresses that gap. I developed a machine learning model for wildfire burn severity classification that integrates satellite-derived vegetation stress indicators with atmospheric reanalysis data across Australia from 2015 to 2024. Specifically, the model uses vegetation optical depth from NASA’s SMAP SPL3SMP_E Version 6 dataset to classify individual fire events as low or high severity based on fire radiative power (FRP) [3].

In this paper, I created a VOD-integrated XGBoost wildfire severity classifier, a binary classification system combining multi-temporal vegetation moisture signals with ERA5 atmospheric covariates. Our model achieves 68% overall accuracy and 0.82 recall on high-severity fires across a test set. The primary contribution of this work is demonstrating that pre-fire vegetation moisture is a stronger predictor of burn severity than same-day atmospheric conditions, indicating that satellite vegetation monitoring could meaningfully extend the forecast horizon for operational wildfire risk systems.

The remainder of this paper is organized as follows. Section 2 details the procedure and experimental design. Section 3 presents results. Section 4 offers conclusions and future work.

2 Procedure

2.1 Software and Environment

All data extraction, preprocessing, and model training were conducted in Python 3.11 using Google Colab as the execution environment. Google Earth Engine (GEE) served as the primary platform for accessing and processing geospatial datasets at scale. The earthengine-api Python client library was used to interface with GEE programmatically. Machine learning was implemented using XGBoost 2.x, with supporting utilities from scikit-learn for preprocessing, evaluation, and dataset splitting. Data manipulation relied on pandas, and results were visualized using matplotlib and seaborn.

All source code is available at Google Colab - Research Project Code.

Component	Detail
Language	Python 3.11
ML Framework	XGBoost 2.x
Key Libraries	scikit-learn, pandas, earthengine-api, seaborn, matplotlib
Platform	Google Colab + Google Earth Engine

2.2 Data

The wildfire data used in this project came from the NASA FIRMS fire archive (M-C61), which was accessed through Google Earth Engine as a FeatureCollection. The dataset included wildfire events across Australia from 2015–2024 and contained information such as the fire acquisition date (`acq_date`), geographic coordinates, and fire radiative power (`frp`). These fire points served as the main dataset for the classification model. NASA’s reporting on increasing wildfire activity and vegetation stress in Australia helped provide background context for the project [1].

Vegetation and soil moisture data came from the NASA SMAP SPL3SMP_E Version 6 dataset [3]. From this dataset, vegetation water content (VOD) and surface soil moisture (`soil_moisture_am`) values were extracted for each fire location. The SMAP dataset has a spatial resolution of about 9 km. To account for missing satellite observations and small timing differences, values were collected within a ± 2 day window around each fire date. In addition to current vegetation conditions, VOD values from one month and three months before each wildfire event were extracted to capture longer-term vegetation stress patterns.

To supplement the vegetation data, atmospheric variables from the ECMWF ERA5-Land hourly dataset were also used through Google Earth Engine. The variables included 2-meter air temperature, total precipitation, 10-meter wind components, and volumetric soil water content. These values were averaged over a ± 3 day window around each fire event at approximately 11 km resolution. These environmental variables were included because previous research has shown that soil moisture and drought conditions are strongly connected to wildfire spread potential [4].

Before training the model, several preprocessing steps were applied. Fire points without valid SMAP coverage (`has_smap == 0`) were removed from the dataset. Missing feature values were filled using the median value of each column. Fire radiative power (FRP) was used to create the binary target variable: fires with FRP values below 20 MW were labeled as low severity, while fires with FRP values greater than or equal to 20 MW were labeled as high severity. Finally, the dataset was split into training and testing sets using an 80/20 stratified split to maintain similar class distributions in both sets.

2.3 Algorithm / Approach

Vegetation Optical Depth (VOD) values were extracted at each wildfire location from the NASA SMAP SPL3SMP_E dataset for three temporal windows corresponding to the fire date, 30 days prior to the fire, and 90 days prior to the fire. There were nine total predictor features that were taken from the ERA5-Land dataset: soil moisture, temperature, precipitation, wind components, and volumetric soil water content. Fire Radiative Power (FRP) values were converted into binary burn severity labels, where $FRP < 20$ was classified as low severity and $FRP \geq 20$ as high severity. An Extreme Gradient Boosting (XGBoost) binary classification model was then trained to predict burn severity. XGBoost iteratively constructs decision trees that minimize classification error through gradient-boosted optimization. Missing feature values were imputed using median feature values prior to training, and the dataset was divided into training, validation, and testing subsets. The classifier was trained using 200

estimators, a maximum tree depth of 4, a learning rate of 0.1, a binary:logistic objective function, and logloss as the evaluation metric.

2.4 Experimental Design

The model was evaluated on a 20% test set using accuracy, per-class recall, confusion matrix (proportional), and XGBoost gain-based feature importances. A three-class version (low / medium / high FRP) was also tested as a secondary experiment. No additional baselines were compared in this phase.

3 Results

3.1 Quantitative Results

	Accuracy	Recall	F1-Score
High	0.69	0.82	0.75
Low	0.64	0.47	0.54

The model achieved particularly strong recall for high-severity fires (0.69), indicating that most dangerous fire events were correctly identified. This is important from a wildfire risk perspective because missing a severe fire can have major ecological and societal consequences. However, recall for low-severity fires was significantly lower at 0.47, meaning that many low-intensity fires were incorrectly classified as high severity.

These results suggest that the model is biased toward predicting high-severity outcomes, most likely because vegetation drought indicators such as VOD are more strongly correlated with extreme fire behavior. While this improves the system’s ability to identify potentially dangerous fires, it also increases the number of false positives.

The three-class confusion matrix shows similar results with a significant portion of the medium fires being misclassified as low; however, a relatively high portion of the high and low fires were classified correctly. This shows that the model is successful in the most extreme cases.

3.2 Figures and Visualizations

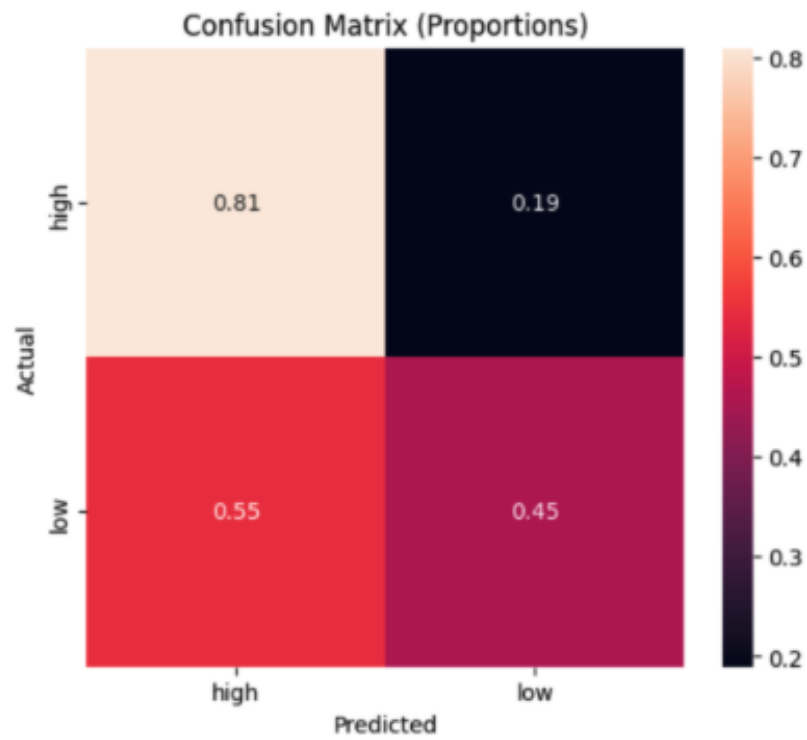


Figure 1: Binary Confusion Matrix

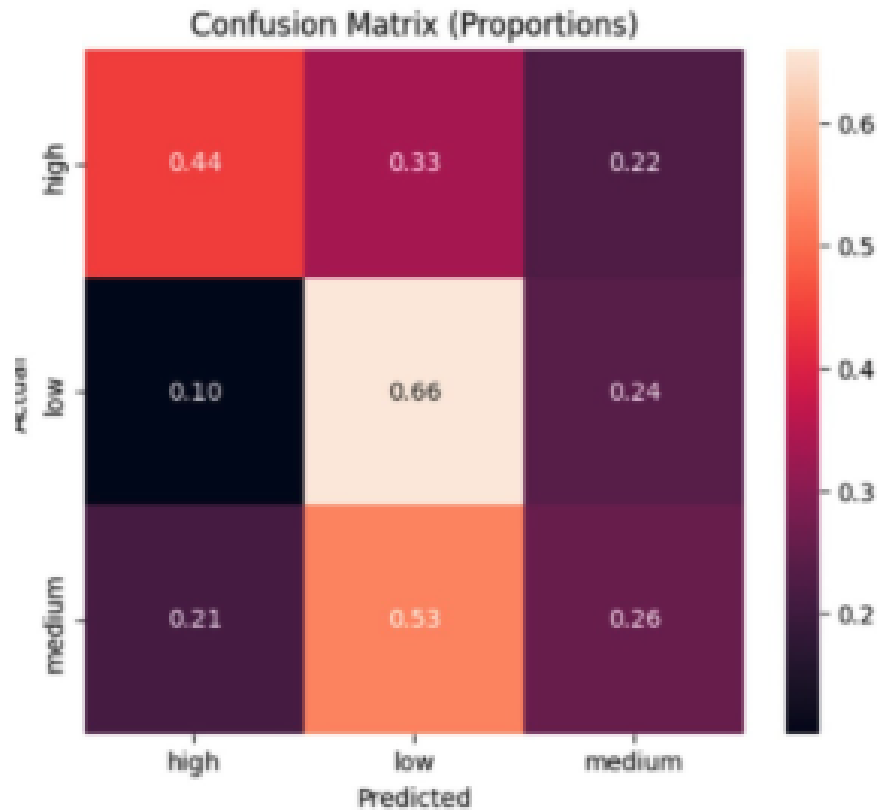


Figure 2: Three Class Confusion Matrix

3.3 Training Curves

The divergence between training and validation loss after approximately 60 boosting rounds suggests moderate overfitting, indicating that the model begins to memorize training-specific patterns beyond this point. Early stopping and additional regularization may improve generalization performance.

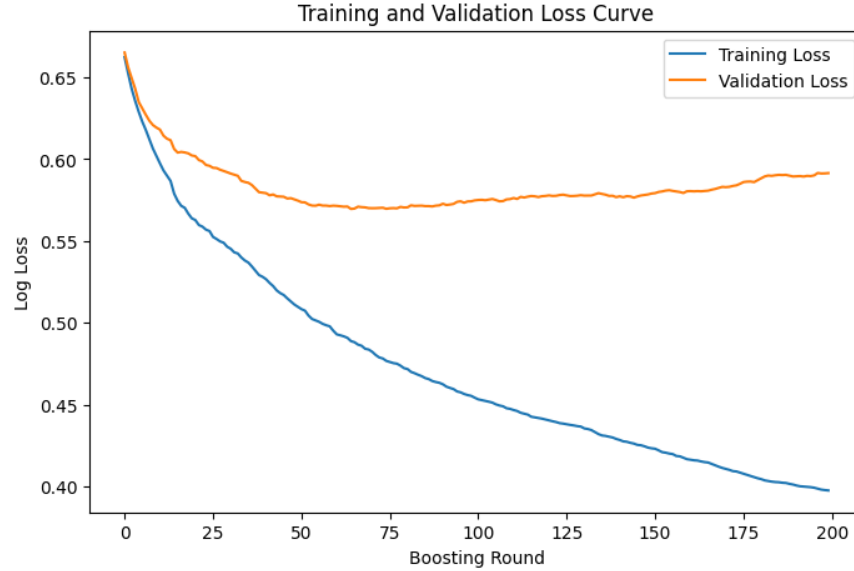


Figure 3: Training Curve

3.4 Final Product

I plan to include an image of my final product once I have it developed. This final project will be a visualization website for my model. This website will be used by firefighters and citizens to see when their area has a high burn severity risk if a fire was to start.

4 Conclusions

In conclusion, this research shows that satellite-derived vegetation water content is a strong predictor of wildfire burn severity in Australia. The XGBoost model achieved approximately 68% binary classification accuracy using nine features derived from NASA SMAP and ERA5-Land datasets, with vegetation optical depth (VOD) measured one month prior before the fire serving as the most influential predictor. These findings suggest that dry vegetation conditions, which can be detected weeks in advance through satellite observations, might provide more predictive value than same-day weather conditions alone.

Several limitations should be considered when interpreting these results. First, the 9 km spatial resolution of SMAP data may not perfectly align with point-scale wildfire locations, potentially reducing prediction precision. In addition, the fire radiative power (FRP) threshold used to classify burn severity represents a simplified approximation of a highly complex ecological phenomenon. Finally, the model has not yet been evaluated against a weather-only baseline model, which will more clearly show the value of this tool.

Based on this model, a web app is being developed to allow firefighting agencies to input a location and date to receive a predicted burn severity estimate alongside a real-time vegetation stress map. This system could support earlier risk assessment and more informed wildfire preparedness strategies.

Future research could focus on improving spatial resolution, incorporating additional environmental variables, and benchmarking performance against alternative modeling approaches. The next step would be to apply this process to other fire prone regions and to see if similar results arise.

References

- [1] NASA, “Australian Wildfires and Climate Change,” 2024.
- [2] Holden, Zachary and Swanson, Alan, “Soil Moisture Is a Stronger Predictor of Forest Fire Spread Potential Than Weather in the u.s. Northern Rocky Mountains,” 2025.
- [3] NASA, “SMAP Enhanced L3 Radiometer Global and Polar Grid Daily 9 Km EASE-Grid Soil Moisture V006,” 2026.
- [4] Copernicus, “ERA5-Land Hourly Data from 1950 to Present,” 2026.